



RIVER VALLEY HIGH SCHOOL

YEAR 6 PRELIMINARY EXAMINATION

CANDIDATE
NAME

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CLASS

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CENTRE
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H1 CHEMISTRY

8873/01

Paper 1 Multiple Choice

20 Sept 2018

1 hour

Additional Materials: Multiple Choice Answer Sheet
Data Booklet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name, class and index number on the Optical Answer Sheet in the spaces provided.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the one you consider correct and record your choice in **soft pencil** on the Optical Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

This document consists of 11 printed pages.

- 1 A mixture of argon and another gas is commonly used during welding. At s.t.p, the density of the gaseous mixture is 1.82 g dm^{-3} .

What is the identity of the other gas in the mixture?
[Density of argon = 1.78 g dm^{-3} at s.t.p.]

- A neon
B nitrogen
C oxygen
D carbon dioxide

- 2 Sodium thiosulfate ($\text{Na}_2\text{S}_2\text{O}_3$) is used in the textile industry to remove an excess of chlorine gas from bleaching processes by reducing it to chloride ions.

If 10 cm^3 of 0.20 mol dm^{-3} of sodium thiosulfate requires 192 cm^3 of chlorine gas for complete reaction at r.t.p., which of the following is a possible formula of the sulfur-containing product?

- A S B SO_2 C HSO_4^- D H_2S

- 3 Which of the following ions would undergo the smallest deflection in an electric field?

- A $^{16}\text{O}_2^+$ B $^{16}\text{O}^{18}\text{O}^+$ C $^{16}\text{O}^{18}\text{O}^{2+}$ D $^{18}\text{O}^{2+}$

- 4 The table below shows the fourth ionisation energies of five consecutive elements in the Periodic Table.

Element	V	W	X	Y	Z
Fourth ionisation energy / kJ mol^{-1}	10450	11710	4350	5030	4580

What is the formula of the bromide of V?

- A VBr B VBr_2 C VBr_3 D VBr_5

- 5 Which of the following does **not** contain a singly occupied orbital?

- A N^{2+} B Br^+ C Sc^{3+} D S^-

- 6 The auto-ionisation of bromine trifluoride is represented by the equation:



Which of the following statements is correct?

- A BrF_3 is planar while BrF_4^- is non-planar.
- B BrF_2^+ is linear while BrF_4^- is tetrahedral in shape.
- C The F–Br–F bond angle in BrF_3 is smaller than that in BrF_2^+ .
- D There are more lone pairs of electrons around the Br atom in BrF_3 than that in BrF_2^+ .

- 7 Dimerisation is described as a process in which two identical molecules combine to give a single product. Some examples of dimers include Al_2Cl_6 , N_2O_4 and $(\text{CH}_3\text{CO}_2\text{H})_2$.

Which of the following statements about the above dimers are correct?

- 1 Hydrogen bonds hold the $\text{CH}_3\text{CO}_2\text{H}$ molecules together in the dimer.
- 2 All the nitrogen-oxygen bonds in N_2O_4 are of equal length.
- 3 The bond angle around each aluminium atom in Al_2Cl_6 is 109.5° .

- A 1 only B 1 and 2 only C 2 and 3 only D 1, 2 and 3

- 8 In an experiment to measure the enthalpy change of neutralisation, 20 cm^3 of aqueous sulphuric acid containing 0.02 mol of H_2SO_4 is placed in a plastic cup of negligible heat capacity.

A 20 cm^3 sample of aqueous sodium hydroxide containing 0.04 mol of NaOH , at the same initial temperature, is added and the temperature rises by 15 K .

If the heat capacity per unit volume of the final solution is $4.2\text{ J K}^{-1}\text{ cm}^{-3}$, what is the enthalpy change of neutralisation?

- | | |
|---|---|
| A $-\left(\frac{20 \times 4.2 \times 15}{0.04 \times 1000}\right)\text{ kJ mol}^{-1}$ | B $-\left(\frac{40 \times 4.2 \times 15}{0.04 \times 1000}\right)\text{ kJ mol}^{-1}$ |
| C $-\left(\frac{20 \times 4.2 \times 15}{0.02 \times 1000}\right)\text{ kJ mol}^{-1}$ | D $-\left(\frac{40 \times 4.2 \times 15}{0.02 \times 1000}\right)\text{ kJ mol}^{-1}$ |

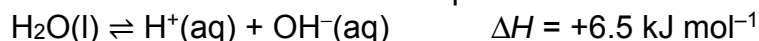
- 9 Given that the following reversible reaction take place at T °C,



What of the following factor affects the K_c value?

- A Decrease in temperature.
 - B Addition of suitable catalyst.
 - C Increment in the concentration of X.
 - D Increment in the concentration of Z.
- 10 How many neutrons is found in 0.180g of Na?
[L = Avogadro constant]
- A 0.00783 L B 0.0860 L C 0.0939 L D 0.180 L

- 11 The auto-ionisation of water is an endothermic process.



Which of the following statements are correct?

- 1 K_w increases as temperature increases.
- 2 pH of solution decreases as temperature increases.
- 3 Acidity of water increases as temperature increases.

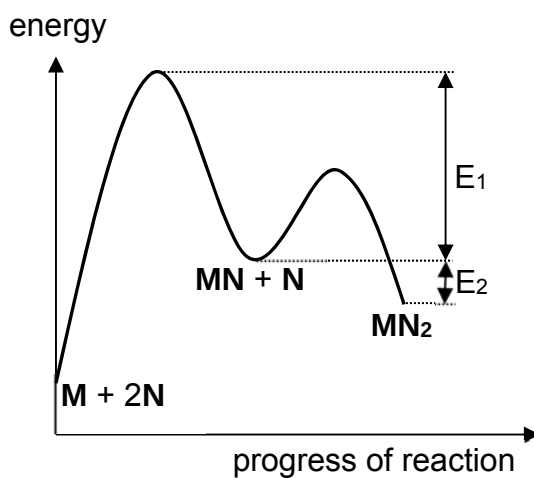
- A 1 and 2 only B 1 and 3 only C 2 and 3 only D 1, 2 and 3

- 12 Use of the Data Booklet is relevant to this question.

Carbon monoxide combust in oxygen to form carbon dioxide only. Calculate the enthalpy change of combustion for this reaction?

- A $-37.0 \text{ kJ mol}^{-1}$ B -155 kJ mol^{-1}
C -285 kJ mol^{-1} D -570 kJ mol^{-1}

- 13 The energy profile diagram of the reversible reaction between **M** and **N** is shown below.

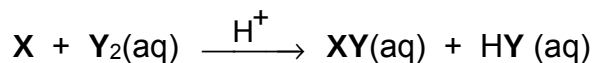


Which of the following statements are correct?

- 1 The activation energy of the reverse reaction is $E_1 + E_2$.
- 2 Rate equation of the reaction is $\text{rate} = k[\text{M}][\text{N}]^2$.
- 3 The equilibrium $[\text{MN}_2]$ increases as temperature increases.

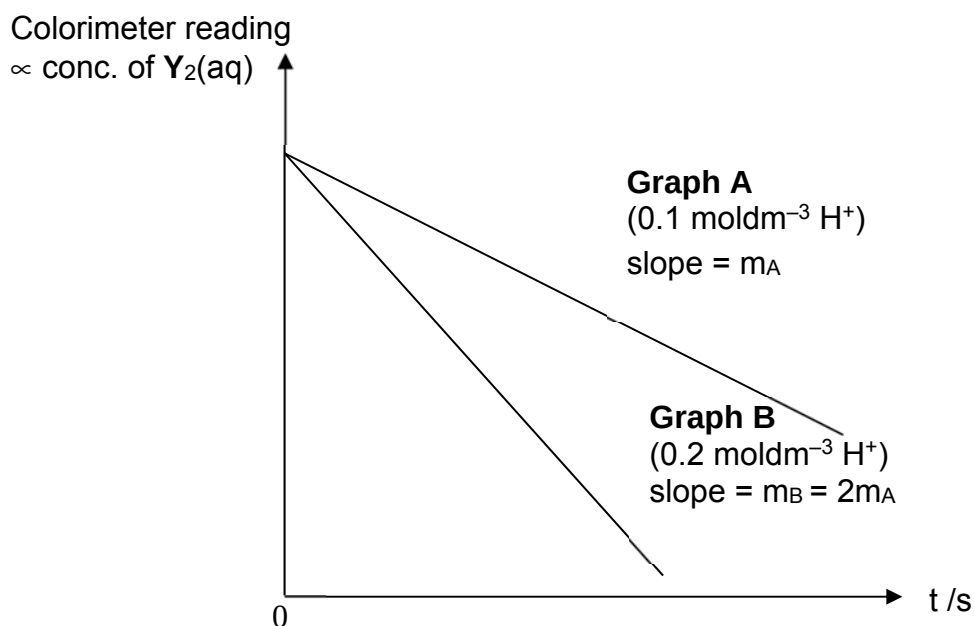
A 1 and 2 only **B** 1 and 3 only **C** 2 and 3 only **D** 1, 2 and 3

- 14 The acid-catalysed reaction between X and Y₂ in aqueous solution has been investigated.



It has been found that rate equation is $\text{rate} = k[\text{X}][\text{Y}_2]^m[\text{H}^+]^n$.

The change in concentration of Y₂ remaining for this reaction is measured by using a photoelectric colorimeter. **Graph A** is obtained when the concentration of hydrogen ions used is 0.1 mol dm⁻³.



What is the overall order of reaction?

- A** 0 **B** 1 **C** 2 **D** 3

- 15** **W** and **X** are 2 different elements in Period 3. The following statements describe some of their properties:

- The oxide of **W** is insoluble in water, but is soluble in both NaOH(aq) and HCl(aq).
- The chloride of **X** dissolves completely in water to give a colourless solution of pH = 2.

Which of the following is the likely identity of element **W** and **X**?

	W	X
A	Al	P
B	Al	Si
C	Mg	P
D	Mg	Si

- 16** A sample of 25.0 cm³ of 1.00 mol dm⁻³ monoprotic acid **HA** (pH = 2.3) is titrated against 1.00 mol dm⁻³ monoprotic base **B** (pH = 14) to find the equivalence point.

Which of the following statement is correct for all acid-base titrations?

- A** The titre value is significantly more than 25.0 cm³.
- B** Energy release per mole of water formed is 57.3 kJ.
- C** Methyl orange is a suitable indicator for the titration.
- D** The degree of dissociation of monoprotic base **B** is 1.

- 17** The shapes of three species, **X**, **Y** and **Z** are bent, trigonal planar and tetrahedral respectively.

Which of the following are possible identities of **X**, **Y** and **Z**?

	X	Y	Z
A	CO ₂	BF ₃	BH ₄ ⁻
B	H ₂ O	PCl ₃	SF ₄
C	NO ₂	NF ₃	XeF ₄
D	H ₂ O	SO ₃	NH ₄ ⁺

- 18 In which of the following pairs of compounds would the first compound have a higher melting point than the second compound?

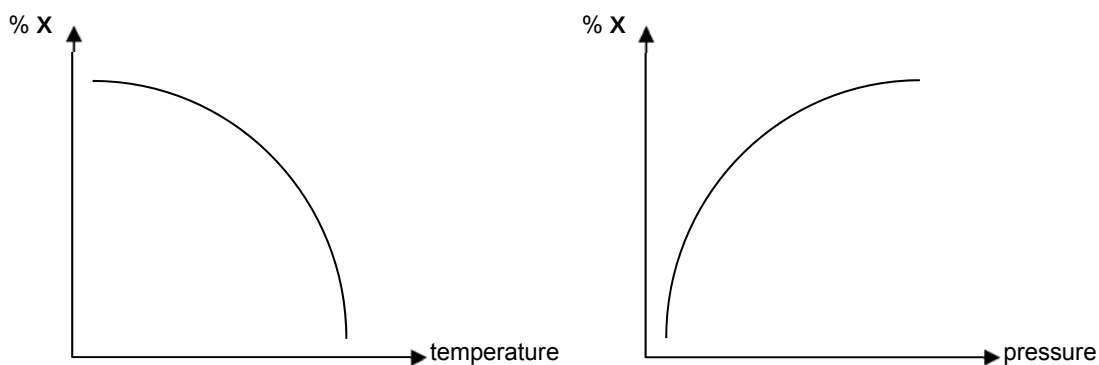
1 AlF_3 AlCl_3

2 KCl NaCl

3 NH_3 H_2O

A 1, 2 and 3 B 1 and 2 only C 2 and 3 only D 1 only

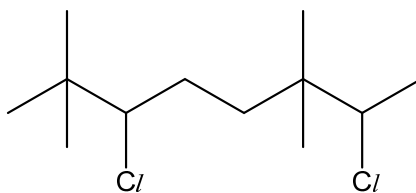
- 19 Compound X is formed during a gas phase reaction. The graphs below show how the percentage of compound X at equilibrium varies with temperature and pressure.



Which of the following responses regarding the equilibrium is correct?

	enthalpy change of reaction	effect of increase in temperature on K_c	effect of increase in pressure on K_c
A	negative	decrease	decrease
B	negative	decrease	unchanged
C	positive	increase	unchanged
D	positive	increase	increase

- 20 The following molecule is a derivative of linalool.



How many stereoisomers may be formed when the molecule is heated under reflux with excess ethanolic potassium hydroxide?

- A 2 B 3 C 4 D 5
- 21 Ethane reacts with chlorine gas in the presence of ultraviolet light to form a mixture via free radical substitution.
Which statement about this reaction is true?
- A Homolytic fission occurs only in the initiation step.
B Bond formation occurs only in the termination step.
C Chloroethane is formed only in the propagation step.
D Small quantities of butane is formed only in the termination step.
- 22 Which property of the Group 1 elements (sodium to caesium) decreases with increasing atomic number?
- 1 the reducing power of the elements
 - 2 the angle of deflection of singly charged cations in an electric field
 - 3 the electropositivity of the elements
- A 1 and 2 only B 1 and 3 only C 2 only D 3 only
- 23 Which reaction represents standard enthalpy change of formation?
- 1 $\text{NH}_3(\text{g}) + \text{HCl}(\text{g}) \rightarrow \text{NH}_4\text{Cl}(\text{s})$
 - 2 $\text{I}_2(\text{g}) + 3\text{Cl}_2(\text{g}) \rightarrow 2\text{ICl}_3(\text{g})$
 - 3 $\text{C}(\text{s}) + \text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g})$
- A 1 only B 2 only C 3 only D 1 and 2 only

24 Which of the following properties increases on descending Group 17?

- A volatility of X_2
- B strength of $X-X$ bond
- C ease of oxidation of X^- ion
- D thermal stability of HX

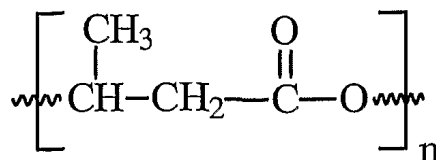
25 Which of the following statement correctly describes low density poly(ethane) (LDPE)?

- A less rigid than high density poly(ethane) (HDPE).
- B lower ductility than high density poly(ethane) (HDPE).
- C lower flexibility than high density poly(ethane) (HDPE).
- D higher tensile strength than high density poly(ethane) (HDPE).

26 Which of the following are possible uses of polyamide, poly(vinyl alcohol) and poly(propene)?

	polyamide	poly(vinyl alcohol)	poly(propene)
A	Clothes	Eye drops	Storage for strong alkali
B	Clothes	Raincoats	Storage for alcohol
C	Wrinkle-free shirt	Eye drops	Storage for alcohol
D	Wrinkle-free shirt	Raincoats	Storage for strong alkali

27 Which of the following statements correctly describes the polymer below?



- 1 The polymer is formed from condensation polymerisation.
- 2 The monomer contains $-\text{COOH}$ and $-\text{OH}$ groups.
- 3 The molecular formula of the polymer is n times that of the monomer.

- A 1, 2 and 3 B 1 and 2 only C 2 and 3 only D 1 only

- 28** Which of the following statements are true for thermoplastic polymers but not true for thermoset polymers?
- 1 The polymer burns upon heating.
 - 2 The polymer can be recycled.
 - 3 There is little or no cross-linkages between polymer chains.
- A** 1, 2 and 3 **B** 1 and 2 only **C** 2 and 3 only **D** 2 only
- 29** Which of the following statements explains why the zig-zag edge is more reactive than the armchair edge?
- 1 A triple bond could be established at armchair edge.
 - 2 On zig-zag edge each outer carbon atom maintain a radical singlet.
 - 3 The planar reconstruction of hexagonal rings to pentagonal or heptagonal.
- A** 1 only **B** 1 and 2 only **C** 2 and 3 only **D** 1, 2 and 3
- 30** Which of the following statements with regards to nanomaterials are correct?
- 1 Nanomaterials are natural, incidental or manufactured.
 - 2 Nanomaterials are all catalysts due to high surface area to volume ratio.
 - 3 Nanomaterial has at least one dimension in the size range 1000 nm.
- A** 1 only **B** 1 and 2 only **C** 2 and 3 only **D** 1, 2 and 3

- End of Paper -